

Raphael Bernardo

Senior Streaming Engineer • *OTT • CTV* •

Android TV • Fire OS

Architect of an OTT framework powering 50+
broadcast brands — 4K multi-DRM, live linear
TV & VOD playback at scale.

LOC Rio de Janeiro • GMT-3

CIT Brazilian + Portuguese (EU) dual citizen

LANG Portuguese (native) • English (fluent)

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§ 01 Summary

PROFILE • POSITIONING

Senior streaming engineer specialising in OTT / CTV platforms. Ten-plus years shipping production media apps across Android TV, Fire OS, webOS and Tizen — including an OTT streaming framework powering 50+ broadcast brands, 4K multi-DRM playback (Widevine, PlayReady, FairPlay), live linear TV & VOD pipelines, and a Kotlin Multiplatform data layer shared across platforms. As an independent contractor I run multiple client engagements in parallel — at any given time only one is full-time, with the rest as part-time concurrent contracts.

Nexstar Media Group, Inc.

Aug 2025 – Present · current

Senior Fire OS / Android TV Engineer · via Nimble.LA · full remote

FULL-TIME CONTRACT

- Architected an enterprise **OTT streaming framework** powering **50+ TV station brands** across Fire TV and Android TV using **MVVM + Clean Architecture** with a plugin-based integration layer for third-party services.
- Built five specialised **Media3 / ExoPlayer** implementations covering VOD, live, and audio playback with HLS adaptive bitrate.
- Shipped DRM-protected playback (**Widevine, PlayReady, FairPlay**) with both server-side and client-side ad insertion.
- Integrated **12+ analytics platforms** (Adobe Analytics, mParticle, Nielsen DCR, ComScore, Permutive, BlueConic) under GDPR / CCPA compliance.
- Delivered TV-optimised auth (passwordless, magic link, QR-code), entitlement & flexible paywall, and zero-downtime feature rollouts across **50+ product flavors** via remote config; built Leanback TV UI with full d-pad navigation.
- Refactored callback chains to reactive Kotlin **SharedFlow**; resolved fragment-lifecycle leaks; wired **Firebase Crashlytics + New Relic** with JUnit, Mockk, Robolectric, Espresso coverage.

Kotlin

Coroutines / Flow

Media3 / ExoPlayer

Dagger 2

Retrofit2

Moshi

Glide

Android Leanback

Jetpack (Nav, ViewModel, Room)

Firebase

Google IMA

Adobe Marketing SDK

mParticle

Nielsen

ComScore

New Relic

Mux Analytics

JUnit · Mockk · Espresso

FreeCast, Inc.

Mar 2021 – Jul 2025 · 4 yrs 5 mo

Android / Android TV / Fire OS Engineer · full remote

LONG-TERM PART-TIME CONTRACT

- Led the **full migration of the legacy Java codebase to Kotlin**; introduced Use-Case + Clean Architecture and improved end-user UX.
- Implemented **HLS / DASH** playback for **live linear TV and VOD** with Widevine DRM-protected streaming; integrated Google Cast and a paid subscription gateway.
- Maintained and developed companion apps on **webOS and Tizen** alongside the Android TV / Fire OS surfaces.
- Integrated **OTA tuner platforms** (HDHomeRun, AirTV, Alticast, PDAQ) with GStreamer-based media pipelines for secure live broadcast over HLS / DASH.
- Owned **DRM, EPG, tuner orchestration, transcoding**, and multi-device streaming end-to-end.
- Shipped on Google Play (*Select TV Android, FreeCast Watch*) and Amazon (*FreeCast Watch Fire OS*).

Kotlin

Jetpack Compose

Hilt

Coroutines / Flow

Media3

Room · DataStore

Retrofit2

Picasso

Firebase Analytics · Crashlytics · FCM

Mixpanel

webOS · Tizen

HDHomeRun · AirTV · Alticast · PDAQ

GStreamer

Widevine DRM

Claro TV+

Apr 2024 – Dec 2024 · 9 mo

Android TV / Fire OS Engineer · full remote

CONCURRENT PART-TIME CONTRACT

- Greenfield Android TV / Fire OS app for one of Brazil's largest pay-TV operators; **4K streaming** via DASH / HLS with multi-DRM.
- Architected the data layer as a shared **Kotlin Multiplatform (KMP)** submodule consumed by Android, webOS and Tizen clients.
- Multi-module **MVVM** architecture; Media3 playback; analytics across Firebase + Google Analytics 4.

Kotlin Multiplatform (KMP)

Kotlin

Jetpack Navigation · ViewModel · Room

Hilt

Media3

Coroutines / Flow

Firebase + GA4

webOS · Tizen

Uber Technologies

May 2021 — May 2022 · 1 yr

Android Engineer · Maps SDK team · via Matchpoint Solutions · full remote

CONCURRENT PART-TIME CONTRACT

- Worked on the **Maps team SDK** module — JNI/C++ bindings between Uber's native Maps engine and the Android client.
- Contributed to Uber's open-source Android architecture framework (**RIBs**); shipped to Uber Rider, Uber Eats and Uber Driver.
- Operated inside a 10K+ engineer codebase using **Buck, Flipper**, custom analytics / logger / crash reporter and a feature-by-module architecture.

Java · Kotlin

C++ · NDK · JNI

Groovy

Python · SQL

RIBs

Buck

CMake

Flipper

Firebase

Invillia

Apr 2020 — Aug 2020 · 5 mo

Android Engineer · full remote

PART-TIME CONTRACT

- Re-modelled the home page of **Casas Bahia** and **Pontofrio** — converted a static product shelf into a fully dynamic, backend-driven shelf system.
- Implemented end-to-end analytics tracking for product clicks and shelf scroll-depth via Firebase Analytics and custom API calls.

Kotlin

MVVM

Dagger 2

LiveData

Retrofit2

Picasso

Firebase (Analytics · FCM · Remote Config · Crashlytics)

Jenkins CI/CD

Super Revendedores

Oct 2018 — Feb 2021 · 2 yrs 5 mo

Android Engineer · full remote

- Migrated the full codebase from Java to Kotlin; implemented **online payment gateway**, virtual showcases and an SSO link to a partner web ordering portal.
- Built scalable asynchronous order processing on **AWS SQS / SNS** for reliable queuing and notification delivery.

Kotlin

AWS (SQS · SNS · Storage)

Firebase (Firestore · Auth · Remote Config · IAM · FCM)

Facebook OAuth

DbFlow

Dagger 2

MVP

Retrofit2

Concrete · an Accenture Company

Dec 2016 — Oct 2018 · 1 yr 11 mo

Jr. Android Engineer

- Built the **Minha CVC** travel app from scratch — **1M+ downloads**; shop geolocation directing customers to physical stores; itinerary feature reducing customer-service call volume.
- Worked on **Getnet**; migrated MVP to MVVM; introduced RxJava2 and LiveData.

Java 8

Kotlin

RxJava2 · LiveData

MVP → MVVM

Dagger 2

Google Maps API

Retrofit2 · Picasso

Firebase

Jenkins CI/CD

PraPagar (fintech)

Jun 2015 — Dec 2016 · 1 yr 7 mo

Jr. Android Engineer · collaborating with US team (MaxxPotential)

- Built mobile payments via **QR code and geolocation**; first Android + Laravel REST experience.

Android

PHP · Laravel 5

REST

OOP · Design Patterns

FASOLTI · Fábrica de Soluções em TI

Sep 2014 — Jan 2015 · 5 mo

Test Developer

INTERNSHIP

- Documented and implemented automated test cases in **Selenium with TestNG**.

IBM Brasil

Jun 2014 – Dec 2014 · 6 mo

Software Engineering Intern · Rio de Janeiro · while pursuing B.S. at Instituto Infnet

- Wrote **SQL queries against IBM DB2** in support of contract and sales teams; produced reporting spreadsheets and Hyperion-based BI artefacts.
- Documented internal processes via SharePoint; first professional exposure to enterprise databases, REST APIs, and Agile workflows.

IBM DB2

SQL

IBM Hyperion

SharePoint

Python

REST APIs

Git

Raphael Bernardo

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§ 03 Selected Personal Work

INDEPENDENT · OPEN SOURCE · RESEARCH

Arquive

Self-hosted personal media archive & streaming server. In-browser **HLS playback** with GPU-accelerated transcoding cache, **IPTV** (M3U + XMLTV EPG + recording), DLNA/UPnP for smart TVs, distributed GPU-fleet transcoding, FAISS-backed face recognition, local-AI captions via Ollama, plus a native **Apple TV (SwiftUI)** client.

PythonReactHLS · GPU Transcoding

FAISS · InsightFaceOllama · Qwen2.5-VLDLNA

M3U · XMLTVSwiftUI tvOS

Multi-Agent Course Creator

Reference implementation of a **multi-agent AI system** on **Google ADK + Agent-to-Agent (A2A) protocol**, deployed on Cloud Run. Four specialised LLM agents in a Researcher → Judge → Content Builder pipeline with session-state propagation, structured Pydantic outputs, and authenticated inter-agent communication.

PythonGoogle ADKA2A ProtocolPydantic

Google Cloud RunLLM Orchestration

Beer Price Calculator

Android app to find the best beer value (price / ml) with history, mini line / bar / pie / heat-map insights, PT / EN localisation, Play Billing subscriptions and coupon-based unlocks. Published on the Play Store.

KotlinJetpack ComposeHiltRoom

Coroutines / FlowPlay BillingMaterial 3

DynamicMusicApp

Android music player with **three pluggable design systems** users switch at runtime — *Material You* (Google), *Neon Wave* (synthwave), and *Organic Flow* (botanical). One Compose codebase, three radically different visual identities — design-system theming treated as a first-class architectural concern.

KotlinJetpack ComposeMaterial 3MVI

Coroutines / FlowHilt

iSpeakPDF

Privacy-by-design web platform converting PDFs to natural-sounding audio entirely in-browser via WebAssembly + Web Workers — no document leaves the device. WCAG-aligned UI, on-device neural TTS with sentence chunking, OCR fallback for scans, subscription tiers and usage limits.

Next.js · ReactTypeScript

WebAssembly · Web WorkersPDF.jsWebGPU

SupabaseStripe

Custom BTC Cache Miner

Research-grade multi-pool Bitcoin SHA-256d **CUDA** miner that tests the same nonce across concurrent Stratum and GBT sources with safe hot-swaps and a VRAM cache (~85% utilisation, dedup, adaptive watermarks). CPU double-SHA-256 verify, crash-safe outbox.

C++ · CUDACMakenlohmann::jsonOpenSSL

Windows · Linux

§ 04 Education

FORMAL · CONTINUING

Instituto Infnet · Rio de Janeiro

B.S. Computer Engineering — software development, mathematical modelling, OS & hardware architecture.

2013 — 2018

Chiswick House School (Malta)

K-12 / High School Diploma.

1997 — 2005

§ 05

Technical Stack

REFERENCE · ATS INDEX	
STREAMING · MEDIA	OTT CTV (Connected TV) Linear TV VOD Live Streaming Media3 / ExoPlayer MediaCodec HLS LL-HLS DASH CMAF H.264 / AVC HEVC / H.265 HDR10+ AAC Hardware-accelerated decoding Widevine PlayReady FairPlay Google IMA SDK SSAI · CSAI GStreamer HDHomeRun · AirTV · Alticast · PDAQ
MOBILE / TV	Android SDK Android TV Fire OS webOS Tizen Android Leanback Jetpack Compose Compose for TV Kotlin Multiplatform (KMP) NDK · JNI
LANGUAGES	Kotlin Java C++ TypeScript JavaScript Python Groovy SQL PHP (Laravel)
ARCHITECTURE	MVVM MVI Unidirectional Data Flow Clean Architecture Use Case Multi-module RIBs MVP Plugin-based integration
JETPACK · REACTIVE	Coroutines · Flow SharedFlow · StateFlow LiveData ViewModel · Navigation Room · DataStore Hilt · Dagger 2 RxJava2
CLOUD · BACKEND	Firebase (Analytics · Crashlytics · FCM · Auth · Firestore · Remote Config · IAM) AWS (SQS · SNS · Storage) Supabase PostgreSQL REST APIs Retrofit2 · Moshi
ANALYTICS	Adobe Analytics Adobe Marketing SDK mParticle Nielsen DCR ComScore Permutive BlueConic New Relic Mux Mixpanel Google Analytics 4
TESTING · BUILD	JUnit Mockk Robolectric Espresso Paparazzi (screenshot) Turbine (Flow testing) Selenium · TestNG Buck Gradle CMake Jenkins CI/CD Flipper
PERFORMANCE · OBSERVABILITY	Quality of Experience (QoE) Time-to-First-Frame (TTFF) Rebuffering rate Startup-time optimisation Systrace Perfetto Android Profiler
COMPLIANCE · ACCESSIBILITY	GDPR CCPA WCAG 2.2 TalkBack · d-pad focus order DRM & Content Security OAuth · SSO
LANGUAGES (SPOKEN)	Portuguese (native) English (fluent)